

Igor Bispo de Moraes Coelho Correia

AI Specialist | Computer Vision & Deep Learning Engineer

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PROFESSIONAL SUMMARY

AI Specialist with 5+ years applying deep learning and computer vision to mission-critical domains — road infrastructure, SAR remote sensing, medical imaging, and automotive safety. Proven track record across the full ML lifecycle: architecture design, training, optimization, and production deployment. MSc in Electrical Engineering (Signal Processing) from the University of Brasília.

TECHNICAL STACK

Deep Learning & CV	PyTorch, TensorFlow, OpenCV, YOLO, Faster R-CNN, U-Net, Transformers, HuggingFace
Languages	Python, C++, C#
Image Processing	SAR imagery, medical imaging (CT/panoramic), embedded video, depth estimation, visual enhancement
MLOps & Infra	Docker, Kubernetes, Apache Airflow, Nagios, Linux on-premise, AWS, GCP
Data	PostgreSQL, SQL Server, Elasticsearch
Tools	CVAT (annotation), Django, Git

WORK EXPERIENCE

Senior AI Specialist FPFtech · Manaus, Brazil (hybrid)	2025 – present
Developed computer vision systems for automotive safety: object detection and classification in compliance with international industry regulations, depth estimation, and visual enhancement. Worked across the full CV pipeline — preprocessing, image quality assessment, deep learning inference, and performance monitoring on resource-constrained embedded hardware.	
Postgraduate Lecturer — Machine Learning & Deep Learning IPOS · Remote	2025 – present
Teaching Machine Learning with Python and Deep Learning with Python in the AI for Civil Construction specialization program. Responsible for curriculum design, course delivery, and assessment.	
AI Analyst National Transportation Confederation (CNT) · Brasília, Brazil	2022 – 2025
Managed the full AI model lifecycle — design, training, and deployment — for the CNT Road Survey , Brazil's largest highway assessment program. Built and maintained detection, classification, and analysis systems covering 40+ road variables (pavement defects, signage, viaducts, guardrails). Stack: Python, C++, C#, TensorFlow, PyTorch, OpenCV, PostgreSQL, SQL Server, Apache Airflow.	
Postgraduate Researcher University of Brasília (UnB) · Brasília, Brazil	2021 – 2025
Curupira project: built a deforestation and wildfire detection system for Sentinel-1 SAR imagery using PyTorch and deep convolutional networks. Gigasistêmica project: designed and trained models for osteoporosis and atheroma detection in CT and dental panoramic images (U-Net, semantic segmentation). Supervised annotation pipelines in CVAT.	
Data Scientist Zerum IT Engineering · Brasília, Brazil	2021 – 2022
Natural language analysis and fiscal text classification for fraud and anomaly detection. Developed LSTM and Transformer models for time-series data. Production classification API with Django. Stack: Python, TensorFlow, HuggingFace, Elasticsearch.	

EDUCATION

MSc in Electrical Engineering — Signal Processing

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University of Brasília (UnB)

2022 – 2024

BSc in Computer Science

University of Brasília (UnB)

2017 – 2021

PUBLICATIONS

Correia, I. B. M. C.; Farias, M. et al. *Comparative Study of Deforestation Detection in Sentinel-1 Scenes of the Amazon Forest*. In: Proceedings of the XLI Brazilian Symposium on Telecommunications and Signal Processing (SBrT2023). DOI: [10.14209/sbrt.2023.1570923825](https://doi.org/10.14209/sbrt.2023.1570923825).

Correia, I. B. M. C. *Deforestation Detection in SAR Images using Deep Neural Networks*. MSc Dissertation — University of Brasília, Brasília, 2024. Available at: repositorio.unb.br/handle/10482/50921.

LANGUAGES

Portuguese

Native

English

Advanced

Spanish

Intermediate